

GEK-9142C

INSTRUCTIONS

3S7501FS100 Through 3S7501FS120

COMPACT

POWER RECTIFIER CUBICLE

for

ALTERREX* EXCITATION SYSTEM

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GENERAL ELECTRIC

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These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to the General Electric Company.

COMPACT
POWER RECTIFIER CUBICLE

INTRODUCTION

The power rectifier takes the AC output voltage of the alternator exciter and rectifies it to produce a DC voltage which is applied to the field of the main generator. Control of the magnitude of this voltage is accomplished through control of the field of the alternator exciter.

RECEIVING, HANDLING AND STORAGE

RECEIVING AND HANDLING

Immediately upon receipt, the equipment should be carefully unpacked to avoid damaging the apparatus. Particular care should be exercised to prevent small parts being mislaid or thrown away in the packing material.

As soon as the equipment is unpacked it should be examined for any damage that might have been sustained in transit. If injury or rough handling is evident, a damage claim shall be filed immediately with the transportation company and the nearest General Electric Sales Office should be notified promptly.

STORAGE

If the equipment is not to be used as soon as it is unpacked, it should be stored in a clean, dry place and protected from accidental damage. Particular care should be exercised to avoid storing the equipment in locations where construction work is in progress.

DESCRIPTION

PHYSICAL ARRANGEMENT

The rectifier compliment consists of a number of three-phase bridge rectifiers as shown in Figure 5 connected in parallel. In most cases so called double bridge rectifiers (two bridge rectifiers in parallel with a single disconnect switch) are used as the basic rectifier unit. In some cases single bridge rectifiers are used. The number of rectifiers and form (single or double bridge) depends upon the power requirements of the generator field.

Each three-phase bridge consists of four assemblies: three rectifier leg assemblies and a header assembly which mounts below the leg assemblies and also serves as the negative DC bus of the rectifier bridge. Each double bridge rectifier circuit consists of two parallel paths per rectifier leg with two diodes in series in each path. There are twenty-four diodes per double bridge and twelve diodes per single bridge. Each of these diodes is mounted on a rectangular copper heat sink through which de-ionized cooling water from the generator cooling system is circulated. The de-ionized water, which is essentially non-conductive, electrically insulates the heat sinks from each other and from ground. The de-ionized water is carried between heat sinks through insulating tubing which also provides electrical insulation between heat sinks. Sections of insulating tubing provide electrical insulation between the cooling system of the rectifier bridge and the metal piping used to supply the de-ionized water to the bridge. Figure 1 shows a portion of a rectifier leg assembly for a double bridge.

A resistor in parallel with a resistor and capacitor in series is connected across each diode. The divider circuit formed by the individual resistors across each diode forces equal division of reverse voltage across the diodes during the negative (blocking) portion of the voltage cycle. The resistor and capacitor connected in series across each diode filter out high frequency inverse voltage spikes which may occur due to power line transients. For ease of mounting, each R-C circuit has been encapsulated in a plotted block.

Fast clearing protective fuses are included in each rectifier's leg assembly. A fuse is connected in series with each path of each positive and each negative leg of the rectifier bridge to clear the leg in the event two diodes in series in the leg fail by shorting.

A neon light is connected across one of the two rectifiers in each leg. The light should light at all times, if generator field voltage is above 160 VDC. At lower voltages, or if a fuse is open the light will go out. The light will also go out if the particular rectifier is shorted.

Each polarity of each diode is wired to terminal boards mounted in the respective rectifier section of the cubicle to provide readily accessible terminals for checking the diodes. Because the diodes in the double bridges are mounted on common heat sinks, resulting in parallel connections of diodes, it may be necessary to disconnect one of the diodes forming the parallel circuit in order to analyze certain diode faults. Refer to section on maintenance for more detailed information regarding checking of diodes.

A Thyrite* resistor is connected across the DC output busses of the power rectifier to protect the diodes from high peak inverse voltages which may occur as a result of abnormal generator operation. The Thyrite resistor is mounted in the Control cubicle, usually in the manual Control section.

CT's are provided in the AC busses near the position where the busses connect to the terminals of the exciter alternator. These CT's furnish the current limit signal for the Control regulator.

INTERLOCKING

Each bridge rectifier is arranged so that it can be

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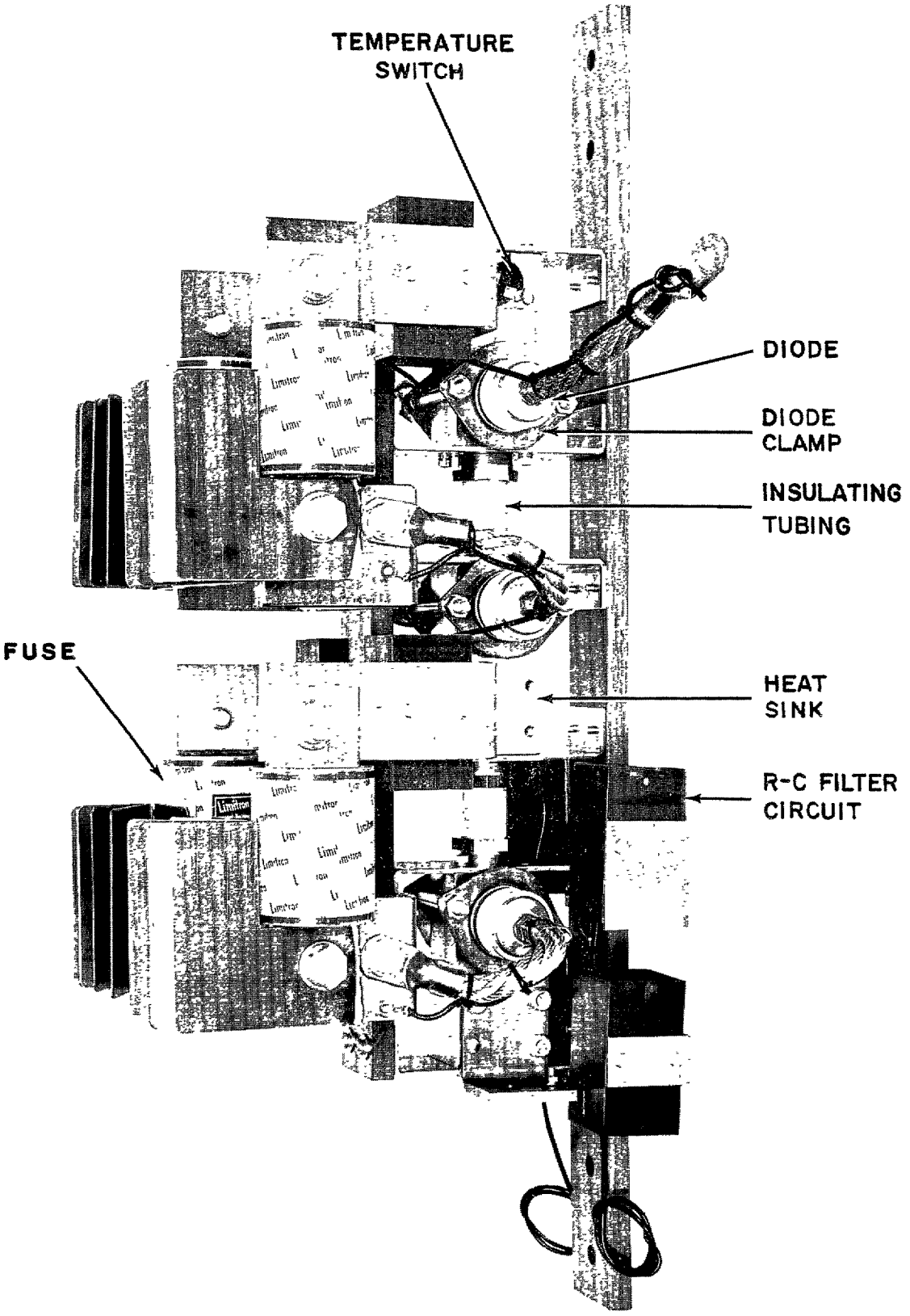

TEMPERATURE SWITCH
DIODE
DIODE CLAMP
INSULATING TUBING
FUSE
HEAT SINK
R-C FILTER CIRCUIT

Figure 1

isolated by opening the disconnect switch associated with the bridge rectifier. (See Figure 2.)

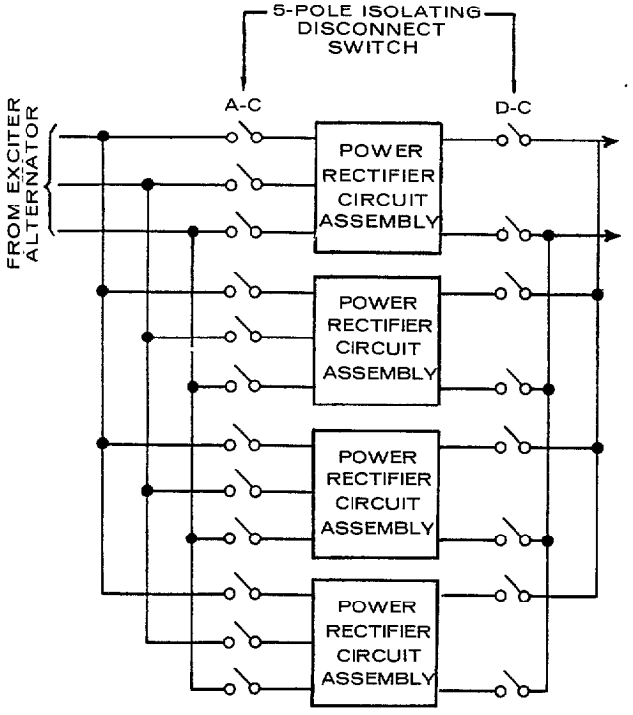


Figure 2

Unless otherwise specified the rectifier disconnect switches are interlocked with key type interlocks to prevent disconnecting too many of the rectifiers at one time and, as a result, causing damage to the remaining rectifiers by overloading. The interlocks and disconnect switches are so constructed that it is necessary to insert the key into the interlock and withdraw the bolt before the disconnect switch can be opened. In the open position the switch mechanism prevents the interlock bolt from being extended until the switch has again been closed. When the bolt is in the withdrawn position, the key is held captive preventing its use on another interlock until the first disconnect switch has been closed. Because of this interlock system only one disconnect switch may be opened at a time and thus only one rectifier may be out of service at a time preventing damage to the remaining rectifiers by overloading.

CAUTION

One key is furnished with each interlock. An operator should have only one key in his possession to be used for all the interlocks to prevent the opening of more than one disconnect switch at a time. Because of this the customer is cautioned to collect all spare keys from the interlocks and store them in a safe place.

SILICON DIODE FUNDAMENTALS

Figure 3 shows the wave-form across one diode when it is operating properly. During the time interval t_1 - t_2, the diode is passing current and the voltage across it is a small amount of forward drop approximately 1 to 1.5 volts. This forward drop varies only slightly as the current changes and does not change with age.

During the interval t_2 - t_3, the diode is blocking current and the wave-shape is shown in Figure 3. The peak of this voltage is equal to the peak of the line to line voltage divided by the number of diodes in the path of the rectifier if the diodes are dividing the reverse voltage equally.

The rectifier should not be hipotted in the field as serious damage to the diodes can occur as a result of the hipot voltage. If it is necessary to megger the power rectifier in the field to check for possible grounds, each diode in the power rectifier must be shorted by a solid connection across it (Cathode to anode) before meggering. De-ionized cooling water of the proper conductivity (see INSTALLATION, COOLING WATER) should be circulating through the power rectifier when the megger test is made.

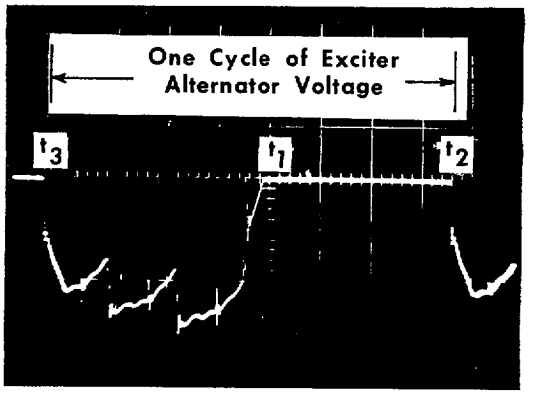


Figure 3

PRINCIPLES OF OPERATION

THEORY

As the theory of three phase bridge rectifiers is well covered in a number of good texts, one of which is entitled "Mercury Arc Power Rectifiers" by A. K. Marti and Harold Winograd, it was felt such theory should not be made a part of this book. The customer is referred to the above text or to any good text with which he may be more familiar for a detailed study of the theory of three-phase bridge rectifiers.

The power rectifier as previously stated in the introduction takes the AC output voltage from the alternator exciter and rectifies it to produce a DC voltage that is applied to the field of the generator. The value of the DC voltage is a function of the magnitude of the AC voltage supplied to the rectifier from the alternator exciter which in turn is controlled by controlling the field of the alternator exciter. Details as to how this is accomplished are covered in separate instruction books (GEK-4798 and GEK-9140) dealing with the control regulator.

SPECIAL CIRCUITS

Each bridge rectifier assembly is provided with an overtemperature switch (TS). This switch is located on the uppermost heat sink of each leg assembly. There are three temperature switches per double or single bridge, one for each of the three leg assemblies of the bridge. The three temperature switches of a bridge are connected in parallel with the control voltage source and the coil of the relay. In case of an overtemperature of any one of the three assemblies making up the bridge, the relay will be energized. One set of contacts on this relay is used to close a circuit to energize a red indicating light for that bridge located on the upper door of the rectifier enclosure section housing the rectifier bridge. A pushbutton is provided on the door for periodic checking of the indicating light. A second set of contacts on the same relay is connected in parallel with similar relay contacts for the other bridges to provide a single annunciator point. In certain installations customer's requirements have dictated a different scheme of relay contact connections. The diagrams furnished with the equipment should always be used for the particular installation.

Any bridge rectifier which indicates an overtemperature should be immediately removed from service by opening the five pole disconnect switch mounted just above the bridge. The reason for the overtemperature condition should then be determined and corrected. Several possible causes are listed below:

1. Low cooling water flow due to low differential water pressure across bridge or an obstruction in the pipe.
2. Poor electrical contact from diode or bus to heat sink.
3. Defective temperature switch.

As previously stated, the contacts of the temperature switch relay may be used to trip an annunciator or may be wired in series with a source of power to sound an alarm. The voltage and current required is not to exceed the relay contact ratings as shown in Table I.

Table I

NON-INDUCTIVE		INDUCTIVE	
VOLTS	AMPS	VOLTS	AMPS
115 AC	30	115 AC	20
230 AC	20	230 AC	15
125 DC	3	125 DC	1
250 DC	0.75	250 DC	0.2

INSTALLATION

LOCATION AND MOUNTING

The power rectifier is supplied in an enclosure designed for mounting adjacent to the alternator exciter. The location of the enclosure should be well ventilated, clean and dry with an ambient temperature not greater than 50°C (122°F). The enclosure is to be mounted so there is adequate clearance around all doors so the equipment is readily accessible.

Detailed information for mounting the power rectifier is supplied on drawings furnished by General Electric, Large Steam Turbine Generator Department. After the enclosure has been mounted and the bus links have been fitted and drilled, the CT's should be installed with the links. The bus fabricated from the shortest bus links and the respective CT should be installed first with the others following in order of length.

COOLING WATER

The cooling water for the power rectifier cubicle is supplied from the de-ionized cooling water system used to cool the generator conductors. This is a high purity water supply which should have a conductivity as measured by a standard conductivity cell, of 0.5 micromhos per centimeter or less. The cooling water supply for the power rectifier must be capable of supplying 5 gallons per minute per double bridge at a pressure differential of 10 PSI gage measured at the inlet and outlet connections to the rectifier cubicle.

CONNECTIONS

During the period of time the power rectifier is being installed, the disconnect switches, which disconnect the individual bridge rectifiers, should be opened to insure that no voltage which could damage the rectifiers is accidentally applied to them.

Connections must be made in accordance with the diagrams supplied for each particular installation. Care must be exercised to determine that the connections are correct to avoid damaging the equipment.

The size of the interconnecting wires, cables, and busses will vary with the rating of the equipment.

Instrument transformers, both current and potential, when used, should be connected as shown in the diagrams.

CAUTION

If the generator or alternator exciter fields are to be hi-potted or meggered in the field, the busses or cables to the power rectifier and the alternator exciter field rectifier must be disconnected to prevent damage to the rectifier diodes.

POLARITY AND PHASE ROTATION

When making connections from the rectifier cubicle to the regulator, polarities and phase rotation should be carefully checked to make certain they are the same as those shown on the applicable diagram.

OPERATION

To operate the power rectifier from start up, the procedure should be as follows. Check to make certain that the inlet and outlet cooling water valves are open at each bridge rectifier and that water is flowing.

Before the generator field breaker is closed, if the system has a generator field breaker, all of the rectifier disconnect switches should be closed. If there is no generator field breaker, the disconnect switches should be closed before AC voltage is applied to the power rectifier from the alternator exciter. This would normally mean before the alternator exciter field breaker is closed or, if a field breaker is not used, before the alternator field rectifier disconnect switch is closed.

CAUTION

When the power rectifier is in operation, no more than one disconnect switch should ever be open at a time to prevent damage to the remaining rectifiers by overloading. The power rectifier can supply the field current required for the generator to carry rated output with one disconnect switch open. (See INTERLOCKING in DESCRIPTION section of Book.)

In case of an overtemperature alarm (See PRINCIPLES OF OPERATION, SPECIAL CIRCUITS), the bridge having the overtemperature condition should be immediately removed from operation by opening its disconnect switch. The cause of the alarm should then be located and the necessary corrections made.

Should a failed diode, either open or shorted, be found in a rectifier as discussed in the MAINTENANCE section of this book, it is not cause for immediate alarm as failures are extremely infrequent. The leg path of the rectifier having the open diode will no longer carry current. This current will be divided among the corresponding rectifier legs of the remaining rectifiers. A shorted diode in the leg of a rectifier will still enable the rectifier to operate at its design current and to withstand required reverse voltage as the remaining diode in the rectifier leg path will withstand the total reverse voltage. Should both diodes in series in a leg path fail by shorting, the protective fuse will blow, opening the leg, and the indicating light will show the disabled leg path. After the diodes have been replaced, it will be necessary to replace the rectifier leg path fuse.

While a failed diode is not cause for immediate alarm, it should be replaced at the earliest convenience as explained in the MAINTENANCE section. Each bridge rectifier is designed to be serviced while the remaining bridges are in operation. The bridge containing the failed diode should be isolated by opening the disconnect switch associated with the particular bridge rectifier. This disconnects both the main AC and DC power busses from the bridge.

MAINTENANCE

The power rectifier should be kept relatively clean and dry. If vibration is present, all screw type connections should be checked regularly to determine that they are properly tightened.

Silicon diodes are used in the rectifiers of the power rectifier cubicle. These diodes are not known to age; therefore, they are either good or faulty and if faulty they should be removed. Diodes can fail either by open circuit or short circuit.

It is recommended that the diodes be checked at the following intervals:

1. At start up.
2. At one month intervals after start up during first year.
3. Yearly after 2 above.
4. After any abnormal system or operating occurrence if there is reason to believe overloading occurred and protective circuits provided may not have operated properly.

There are a number of ways to check diodes. The General Electric Co., Specialty Control Department makes equipment, 3S7932MD116 and 44C215817, to check the diodes with the rectifier in operation.

Diodes can be checked with a cathode ray oscilloscope if available. For the check at startup it is suggested that the power rectifier be operated to give rated no load generator field voltage. With the ground lead of the scope connected to the cathode of the diode and the signal lead of the scope connected to the anode, a good diode should have a wave shape as shown in Figure 3, SILICON DIODE FUNDAMENTALS, DESCRIPTION Section of this book. In order to facilitate connections to the terminals of the diodes for testing, test points for each bridge rectifier are provided on a terminal board which is located in the respective section of the rectifier cubicle. Refer to the diagrams supplied with the equipment for proper connections. During the positive half cycle of the AC voltage while the diode is conducting, the forward voltage drop of approximately 1 to 1.5 volts appears as a straight line, time interval $t_1 - t_2$. During the negative portion of the cycle, the diode is blocking the flow of current and the voltage across the diode appears as shown in the interval $t_2 - t_3$. The diodes in a rectifier leg path (diodes in series

from an AC bus to a positive or negative DC bus) divide the AC voltage applied so that the peak reverse voltage seen on the scope across a given diode will be the peak AC voltage divided by the number of diodes per leg path.

If a diode is shorted, it will not hold off the reverse voltage so will not exhibit the negative portion of the wave form from t_2 - t_3. A diode, when shorted, will usually, though not always, have a small voltage drop across it in both the forward and the reverse directions due to the current flow through the low resistance of the short.

Because the diodes in each double bridge rectifier are mounted on common heat sinks, a number of common circuit points result within each leg of the rectifier. This places pairs of diodes within each leg in parallel (one in each leg path). When a faulty diode is located as a result of a maintenance check, it will be necessary to disconnect one of the parallel diodes in order to determine if one, or both, of the diodes is defective. This is particularly true where the fault condition is a shorted diode (open diodes cannot usually be detected with a scope as explained in the following paragraph). Power should be removed from the rectifier bridge by opening its disconnect switch before attempting to disconnect the pigtail connection of the diode. The disconnected pigtail of the diode should be insulated so it cannot accidentally contact ground or other circuits and then power should be restored to the bridge. If, after the first diode has been disconnected, in the case of a shorted diode, the scope connected at the test points on the terminal board indicates the shorted diode condition has been removed, the first diode was shorted. If the scope still indicates a shorted diode condition, the second diode is shorted and it will be necessary to reconnect the first diode and disconnect the second diode to determine if the first diode may also be shorted.

If a diode is open in a leg, it usually cannot be detected with a scope as the diodes in the corresponding rectifier legs, in parallel with the open diode and the divider action of the resistor capacitor circuits across the diode in each leg cause the forward drop across the open diode to appear very nearly the same as for a normal diode. The R-C circuits and the remaining diodes in the leg tend to share the reverse voltage so the reverse voltage across the diode is practically the same as for a good diode again making it practically impossible to detect an open diode when operated in parallel with other diodes by checking the reverse voltage with a scope.

To locate an open diode while the rectifier is in service, a clamp-on ammeter may in turn be clamped on the pigtail of each diode. An open diode will give a zero current flow indication on the meter. Because of the common circuit points within each rectifier leg, a no current flow indication on two parallel diodes in a rectifier leg does not necessarily mean that these diodes are open as it could also mean that the two remaining diodes in the leg are open.

If such meter indications are obtained, power should be removed from the bridge by opening its disconnect switch. If the bridge has current balancing CT's, the AC and DC busses should be shorted below the switch as explained in the OPERATION Section of this book. Disconnect one of each pair of parallel diodes in the rectifier leg. With the voltage removed from the rectifiers, use an ohmmeter to measure the forward and reverse resistance of each diode in the leg. For a good diode the forward resistance should be very low compared to the reverse resistance. No exact values can be given as different meters use different battery voltages and will, therefore, give different readings. The best way to obtain a calibration of what the forward and reverse resistance should be for a given meter is to actually measure these values on a diode known to be good. An open diode will have a high forward resistance as well as a high reverse resistance. A shorted diode will have practically zero forward and reverse resistance.

Bad diodes may also be found with the equipment in service by checking with a DC voltmeter. The same requirements for disconnecting parallel diodes within a rectifier leg apply when checking with a voltmeter as applied for checking with a cathode ray oscilloscope. The voltmeter should have an internal resistance of at least 1000 ohms per volt and its range should be selected in the following manner. Measure the AC line to line voltage to the rectifier with an AC voltmeter. Multiply this value by $\sqrt{2}$ to determine the peak value. The DC voltmeter should then be selected to have a full scale reading at least equal to this peak value. If the negative lead of the voltmeter is connected to the anode of the diode and the positive lead is connected to the cathode of the diode, the meter will read up scale in proportion to the reverse voltage held off by the diode. The voltmeter will give essentially this same indication for a good diode as an open diode when operated in parallel with other diodes as in the leg circuits of the power rectifier. A shorted diode will have a very low and approximately equal forward and reverse voltage drop so the DC voltmeter should read nearly zero.

An open diode cannot usually be detected with a voltmeter for the same reasons as given for not being able to detect it with a scope. To check for an open diode with the rectifier in service, a clamp-on ammeter should be used as previously explained. An ohmmeter may also be used but power must, of course, be removed from the rectifier before making this test.

Individual rectifiers can also be checked as shown in Figure 4. With the switch in position 1, the ammeter should read approximately 12 amperes. With switch in position 2, the ammeter should read zero. The DC source should be a battery rather than a rotating exciter since the latter may have voltage spikes that may damage the diode.

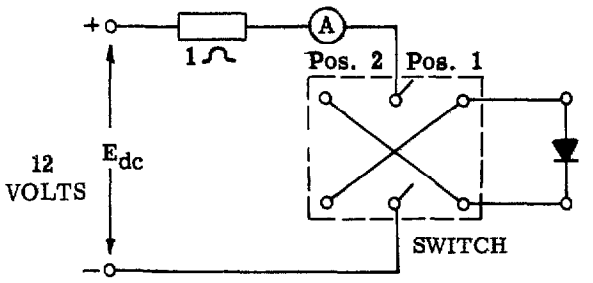


Figure 4

When replacing a diode, it should always be replaced with the same catalog number and grade of diode as removed. For example an A296PEY45, grade 5 diode should be replaced with an A296PEY45, grade 5, and not with any other A296 diode or grade.

If it becomes necessary to replace a diode, the following procedure should be followed:

1. Isolate the rectifier with the faulty diode by opening the disconnect switch associated with the rectifier. Jumper AC and DC busses to rectifier together below switch with #8 or larger wire and ground. (Jumpering and grounding not necessary for power rectifiers which do not have balancing CT's.)

2. Disconnect diode pigtail from bus. Remove diode by loosening and removing bolts which hold diode clamp. In some locations, two diode clamps, holding diodes on opposite sides of the heat sink, are mounted with the same set of bolts.

3. Remove old dielectric compound from the surface of the heat sink. A cloth saturated with isopropyl alcohol will aid in removing the old compound. Where two diodes are clamped on opposite sides of the heat sink with one set of bolts, the old compound should be removed from both sides of the heat sink, which will mate with the diodes, as well as from the surface of the old diode which will be returned to service.

4. Before installing a new diode or returning an old diode to service, the surface of the diode which mates with the heat sink should be lightly coated with a dielectric compound such as Wakefield Type 120, produced by Wakefield Engineering Corp., Wakefield, Massachusetts, or General Electric Company compound G-640 available from General Electric Company, Silicon Products Dept., Waterford, N. Y.

5. Install diode with clamp and bolts. Before the bolts are tightened, the diode should be rotated within the clamp to align the pigtail terminal with the bus to which it is to be connected. This is necessary to avoid unnecessary stress in the pigtail which in turn could damage the diode. The clamp bolts should be tightened evenly until the clamp is drawn flat. The pigtail terminal should then be bolted to its bus connection.

When replacing faulty diodes a sufficient number of rectifier bridges must always be left connected to supply the generator field current. No more than one rectifier bridge should be disconnected at a time.

If the power rectifier is to be out of service with the ambient temperature below freezing the water should be drained from the heat sinks and piping. A drain plug is provided in the front side of each rectifier bridge header (large heat sink at the bottom of the rectifier bridge) and at various points in the rectifier piping for this purpose. Care should be taken to replace drain plugs and restore water flow before the rectifier is again energized.

RENEWAL PARTS

When ordering renewal parts, address the nearest General Electric Sales Office; specify the quantity required and give the catalogue number or describe the required part in detail. Also give the complete nameplate rating of the equipment.

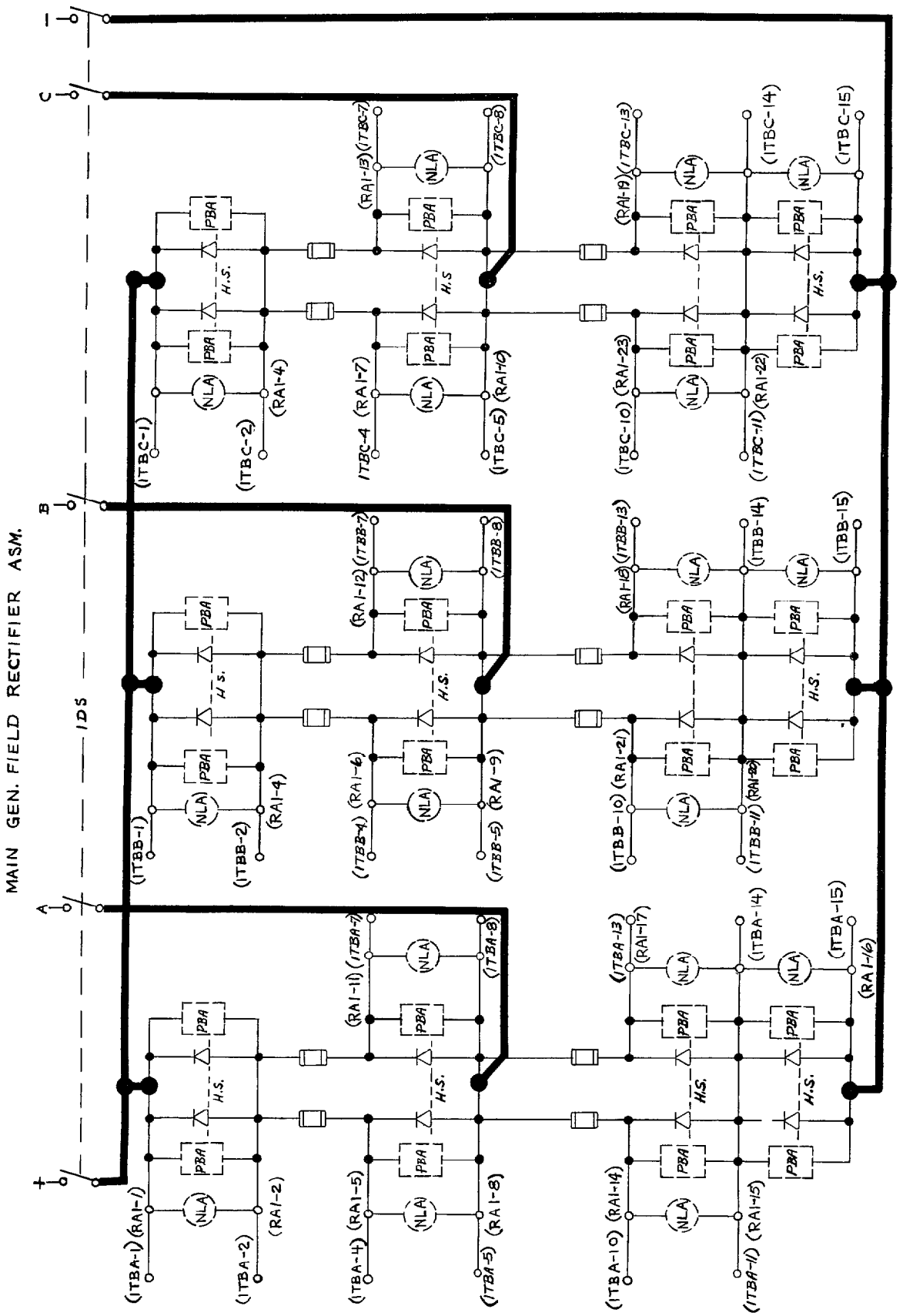

MAIN GEN. FIELD RECTIFIER ASM.
IDS
+
A
B
C
-
(ITBA-1) (RAI-1)
(ITBA-2)
(RAI-2)
(ITBA-4) (RAI-5)
(ITBA-5) (RAI-8)
(ITBA-7) (RAI-11)
(ITBA-8)
(ITBA-10) (RAI-14)
(ITBA-11) (RAI-15)
(ITBA-13) (RAI-17)
(ITBA-14)
(ITBA-15) (RAI-16)
(ITBB-1)
(ITBB-2)
(RAI-4)
(ITBB-4) (RAI-6)
(ITBB-5) (RAI-9)
(ITBB-7) (RAI-12)
(ITBB-8)
(ITBB-10) (RAI-21)
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(RAI-19) (ITBC-13)
(ITBC-14)
(ITBC-15)
PBA
NLA
H.S.

Figure 5